

Machine Learning Specialization

Welcome

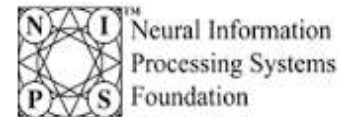
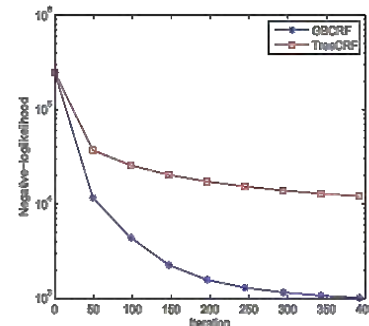
Emily Fox & Carlos Guestrin

Machine Learning Specialization

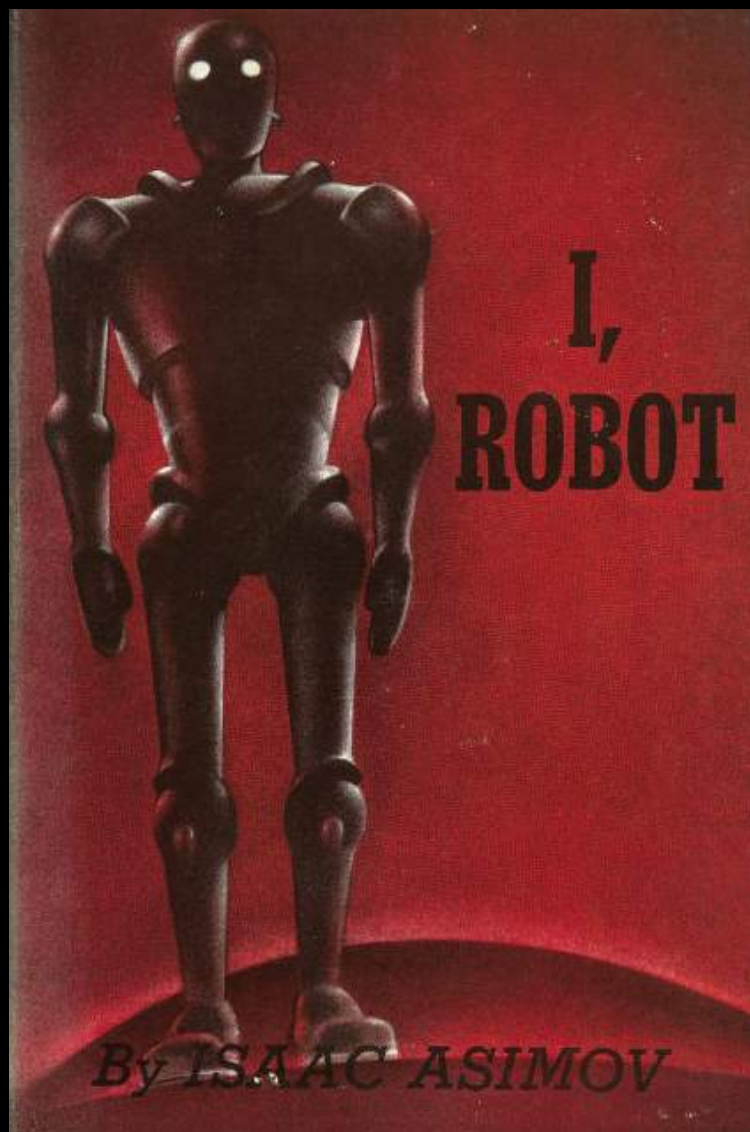
University of Washington

Machine learning is
changing the world

Old view of ML



ICML





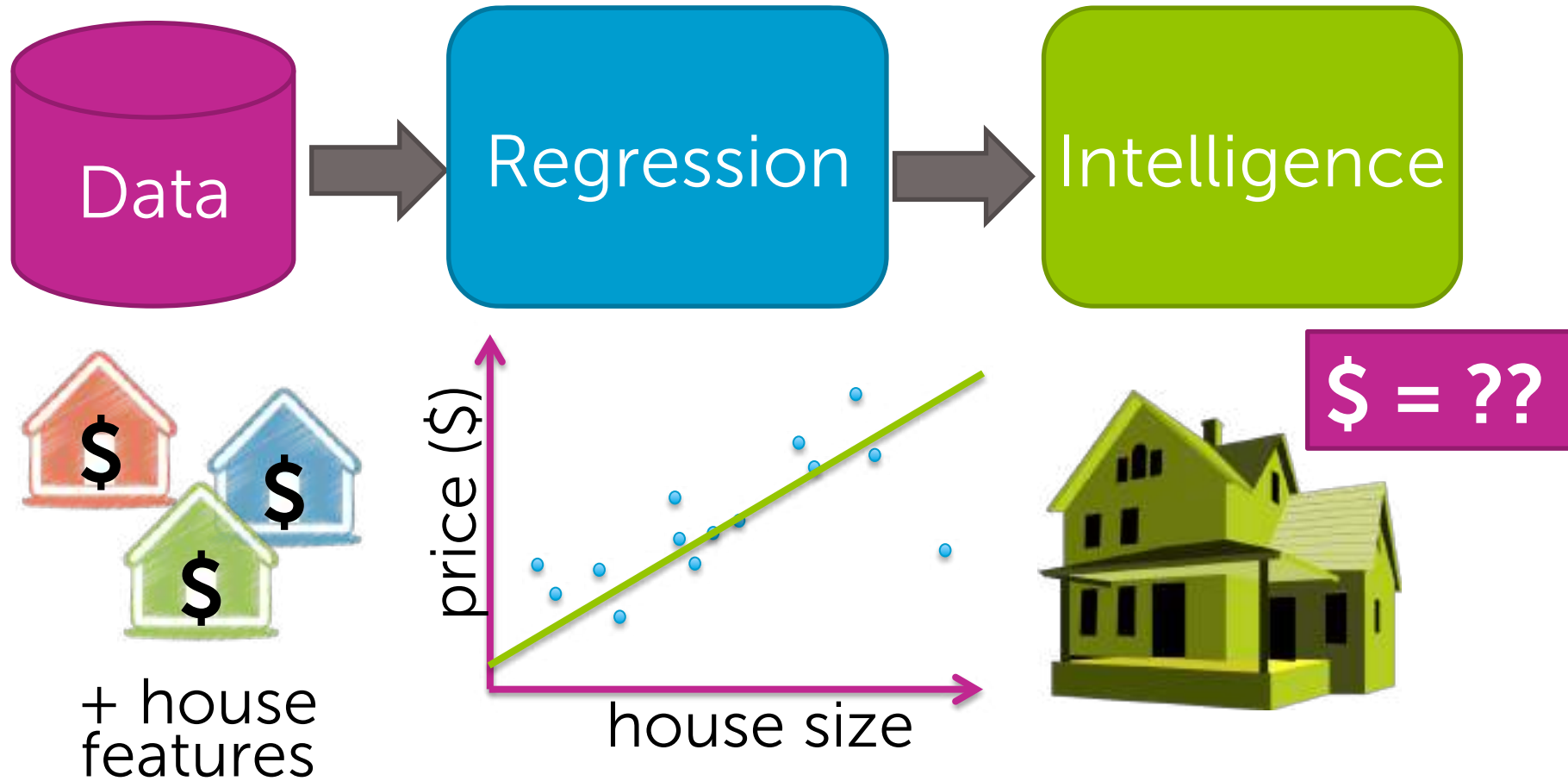
The machine learning pipeline



ML case studies

Case Study 1:

Predicting house prices

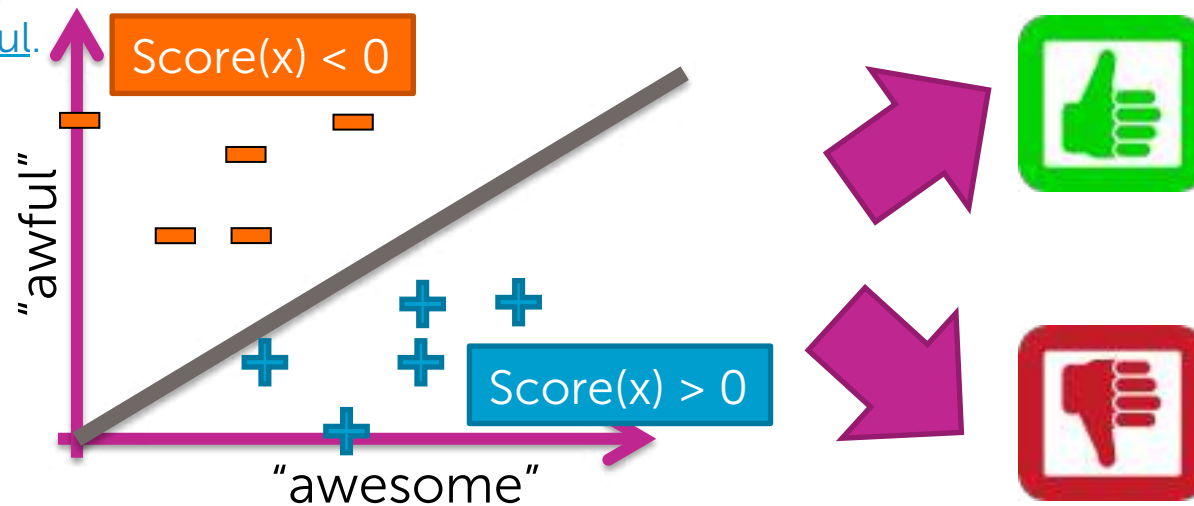


Case Study 2: Sentiment analysis



Sushi was awesome,
the food was awesome,
but the service was awful.

All reviews:



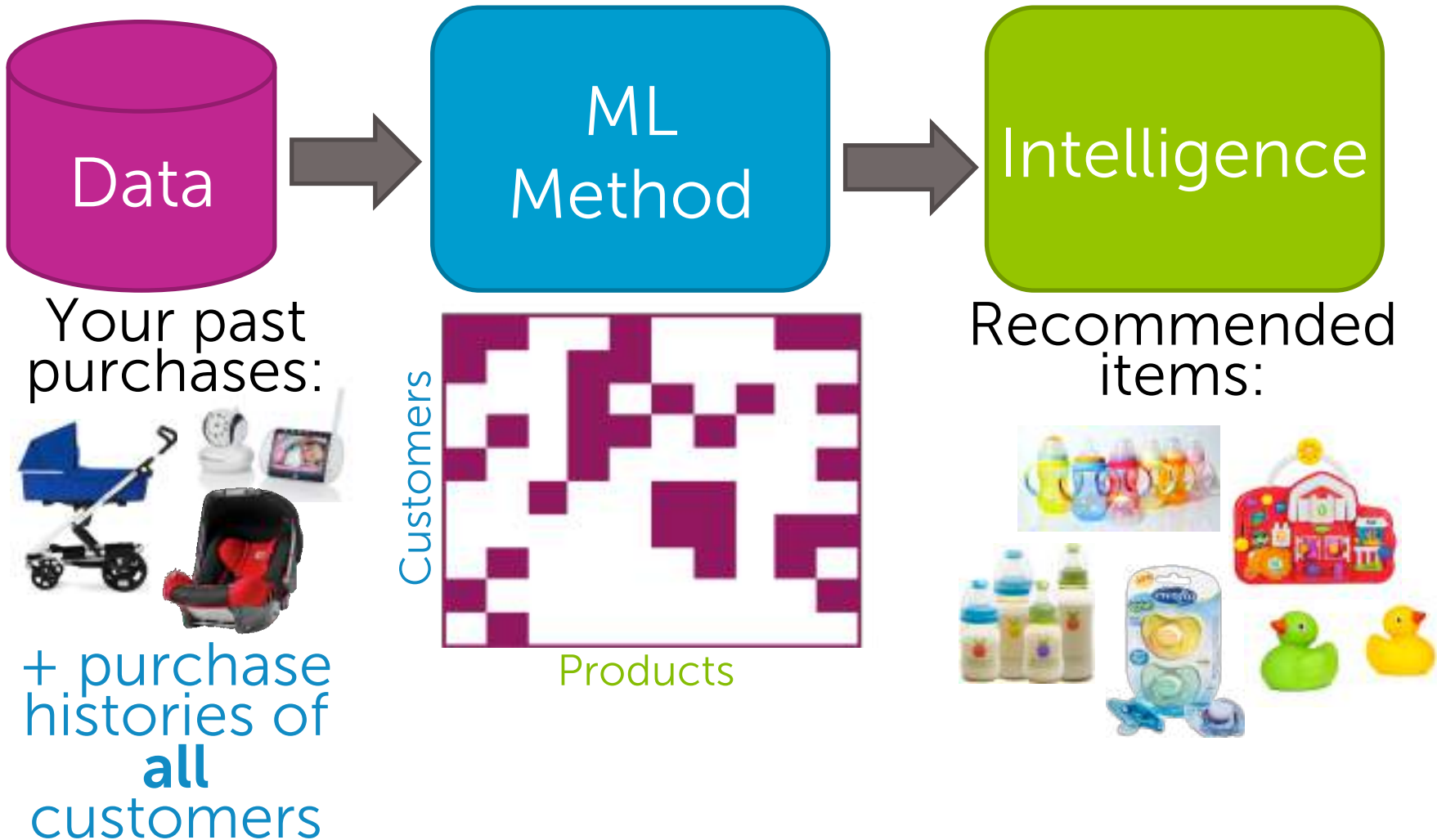
Case Study 3:

Document retrieval



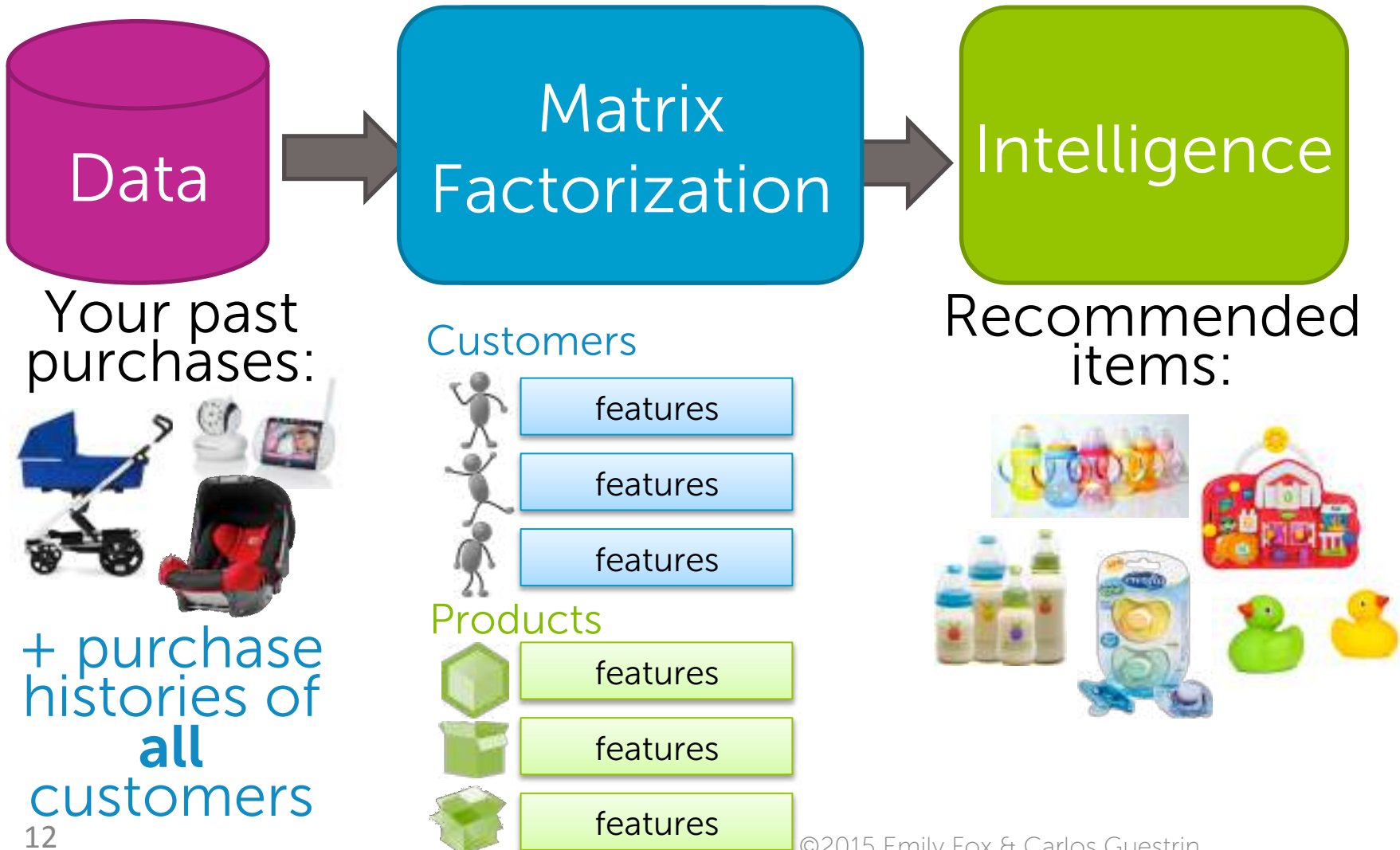
Case Study 4:

Product recommendation



Case Study 4:

Product recommendation

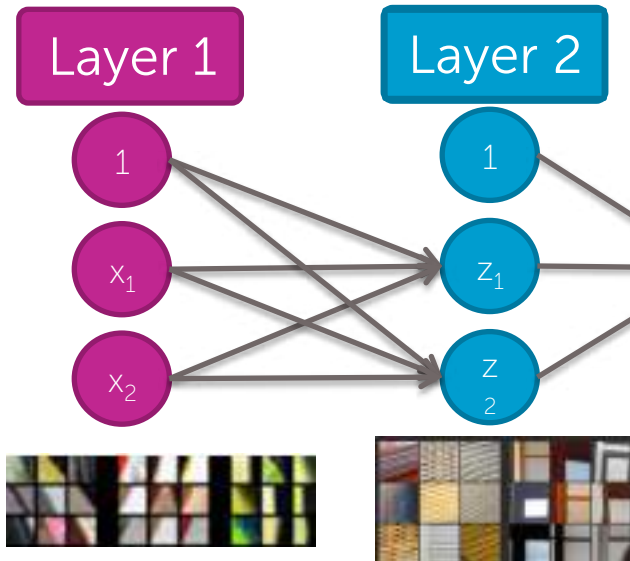


Case Study 5:

Visual product recommender



Input images:



Nearest neighbors:

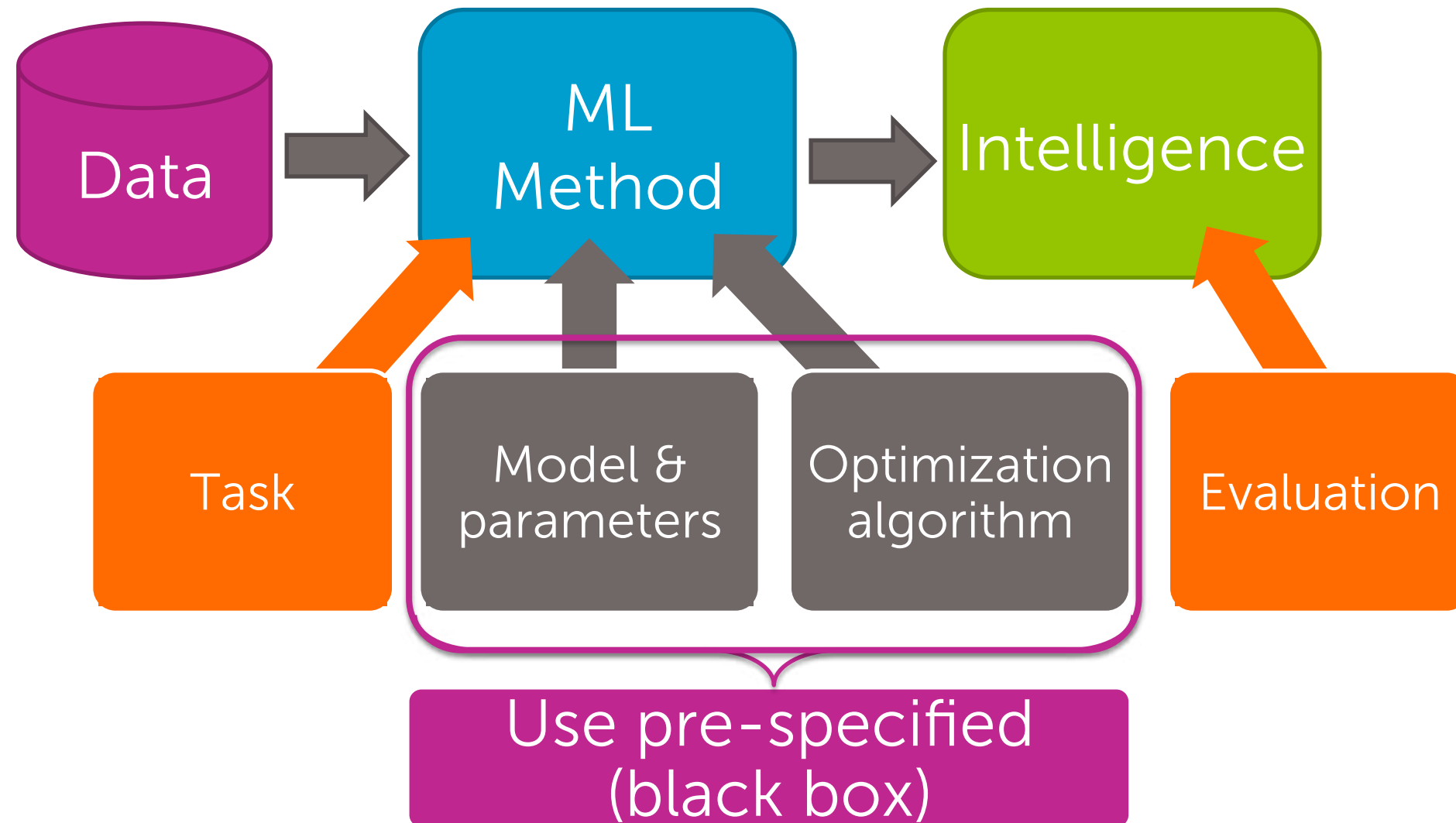


A unique ML specialization

Not like other ML courses out there...

From use cases to
models & algorithms

First course is about building, evaluating and deploying *intelligence in each case study*...



Subsequent courses provide depth in models & algorithms, but still use case studies

2. Regression
3. Classification
4. Clustering & Retrieval
5. Matrix Factorization & Dimensionality Reduction
6. *Capstone*: Build an Intelligent Application with Deep Learning

2. Regression

Case study: Predicting house prices

Models

- Linear regression
- Regularization: Ridge (L2), Lasso (L1)

Algorithms

- Gradient descent
- Coordinate descent

Concepts

- Loss functions, bias-variance tradeoff, cross-validation, sparsity, overfitting, model selection

3. Classification

Case study: Analyzing sentiment

Models

- Linear classifiers (logistic regression, SVMs, perceptron)
- Kernels
- Decision trees

Algorithms

- Stochastic gradient descent
- Boosting

Concepts

- Decision boundaries, MLE, ensemble methods, random forests, CART, online learning

4. Clustering & Retrieval

Case study: Finding documents

Models

- Nearest neighbors
- Clustering, mixtures of Gaussians
- Latent Dirichlet allocation (LDA)

Algorithms

- KD-trees, locality-sensitive hashing (LSH)
- K-means
- Expectation-maximization (EM)

Concepts

- Distance metrics, approximation algorithms, hashing, sampling algorithms, scaling up with map-reduce

5. Matrix Factorization & Dimensionality Reduction

Case study: Recommending Products

Models

- Collaborative filtering
- Matrix factorization
- PCA

Algorithms

- Coordinate descent
- Eigen decomposition
- SVD

Concepts

- Matrix completion, eigenvalues, random projections, cold-start problem, diversity, scaling up

6. Capstone:

An intelligent application using deep learning

Build & deploy
a recommender using
product images and
text sentiment

This specialization is for you if...

Level of the specialization

Motto:

*tough concepts made intuitive
and applicable*

minimize prereq knowledge

maximize ability to develop and deploy

learn concepts through case studies

Target audience



Software engineer



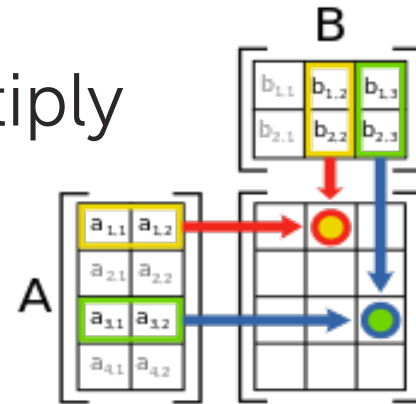
Scientist



Data enthusiast

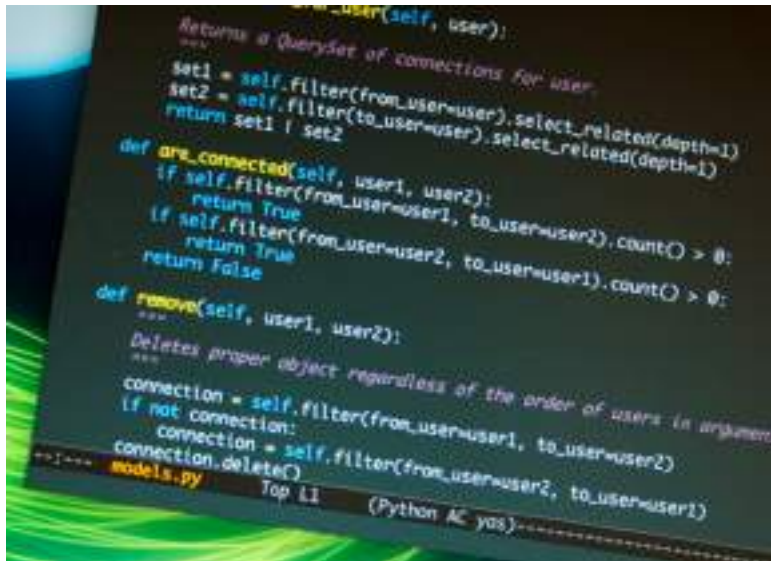
Math background

- Basic calculus
 - Concept of derivatives
- Basic linear algebra
 - Vectors
 - Matrices
 - Matrix multiply



Programming experience

- Basic Python used
 - Can pick up along the way if knowledge of other language



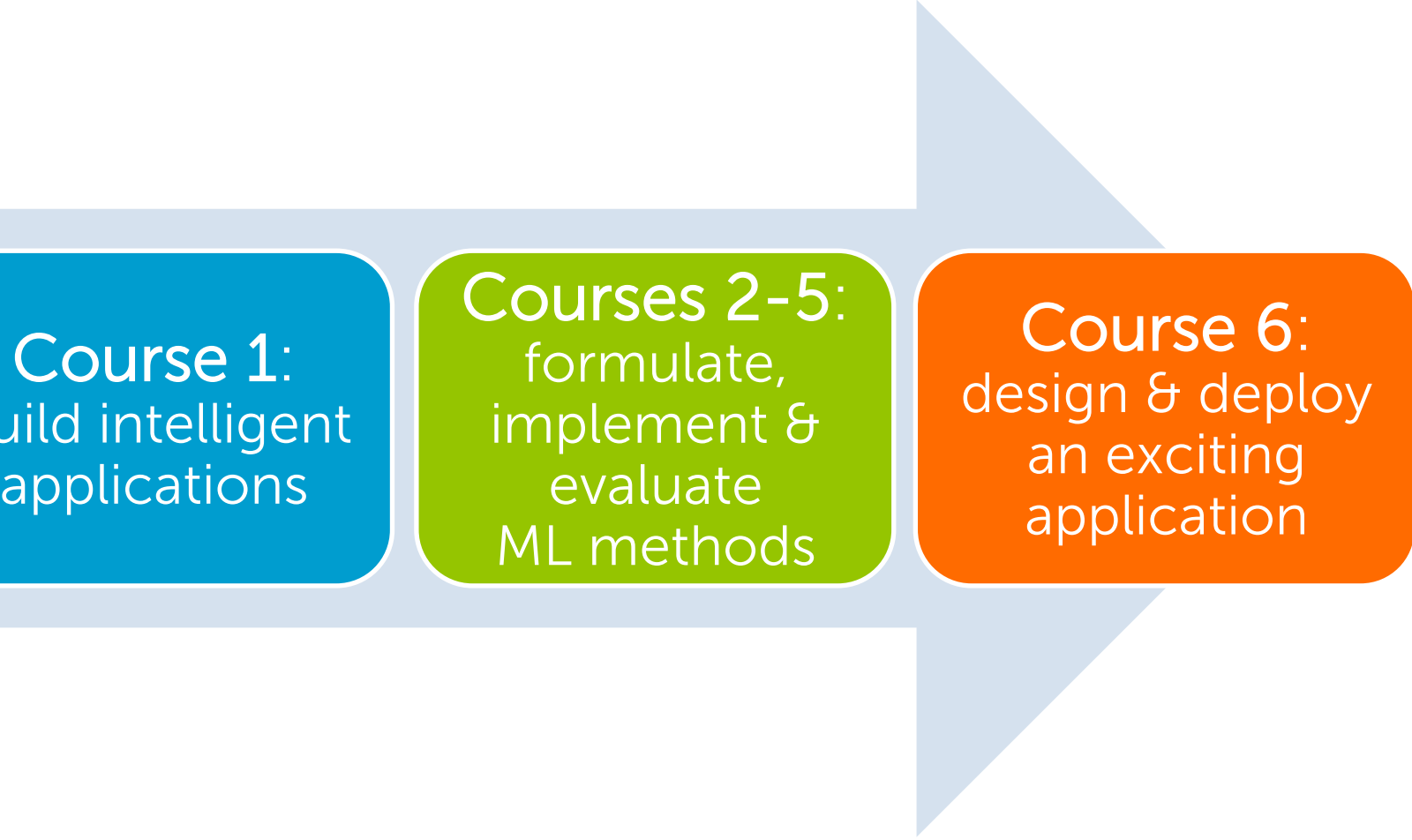
Computing needs

- Basic desktop or laptop
- Access to internet
- Ability to:
 - Install and run Python
 - Store a few GB of data



You'll be able to do
amazing things...

Our journey together...



Course 1:
build intelligent
applications

Courses 2-5:
formulate,
implement &
evaluate
ML methods

Course 6:
design & deploy
an exciting
application

The Capstone Project:

Build and deploy an intelligent application with deep learning

An intelligent recommender using images & text

We will do something even more exciting...

